

1) HISTORY PREFIX INPUT

History prefix h_t^-

o_1	Observation (decision-relevant cue)	Lang ₁ “go to the key”
a_1	Action	a_1 (left)
r_1	Reward	r_1
Lang ₁	Language cue	Lang ₁ “go to the key”

Current observation o_t

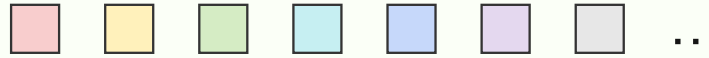
Encoder
 E_ϕ

2) DISCRETE MEMORY BOTTLENECK

Discrete code m_t (low-rate persistent)



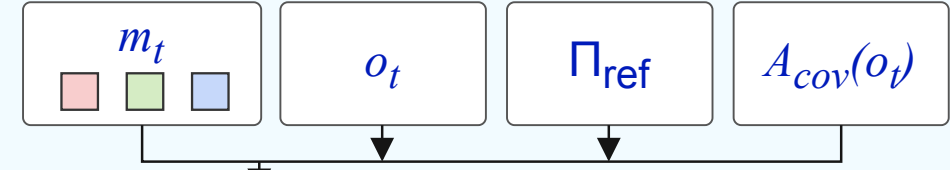
Finite codebook $M = \{m^{(1)}, m^{(2)}, \dots, m^{(K)}\}$



Rate $R(m_t) = -\log_2 P(m_t)$ (bits)

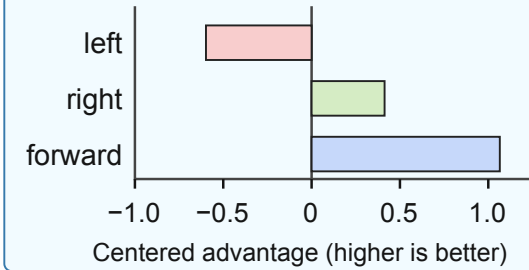
same code => similar centered action advantages

3) DEPLOYMENT



Decoder D_ψ

Centered advantages $\hat{A}^\pi(o_t, a)$



$a_t = \operatorname{argmax}_a$
 $D_\psi(m_t, o_t, \Pi_{ref}, A_{cov})$

Decision a_t

TRAINING ONLY: COUNTERFACTUAL REPLAY-Q

History-consistent
snapshot at time t
(from dataset)

$o_t \ a_{1:t-1} \ r_{1:t-1} \ \text{Lang}_{1:t}$

Fork
candidate
actions

a_1 (left)

a_2 (right)

$\vdots$

a_k (forward)

Frozen
continuation
policy $\pi_{thereafter}$
(frozen)

$G^{(1)}$

$G^{(2)}$

$\vdots$

$G^{(k)}$

Aggregate
returns
(discounted)
 $\hat{Q}^\pi(a_t, o_t, a)$

Centering
 $\hat{Q}^\pi \rightarrow \hat{A}^\pi$
(zero-mean)

Loss (training objective)

Huber loss + $\beta R(m_t)$

→ Supervision (training only)
→ Data flow (deployment & training)

Evidence: Binary
2 codes: 100%

Evidence: Frozen Qwen2.5
150/150 decisions preserved

Evidence: MiniGrid 4-code Q-memory
99.76% on S13, 98.04% on S17

BOUNDARY NOTE: Claim is conditional on the declared policy class Π_{ref} and covered action set A_{cov} ; not universal state sufficiency.